RWANDA NUTRITION

Design and Development of a Database-Driven Web Application Using Django

Project Report

Course: Web Technologies and API Services
Assessment: Practical Mid-Term Project Work

Group 4

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Framework: Python and Django
Frontend: HTML, CSS, JavaScript, Bootstrap and AdminLTE

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Abstract

Rwanda Nutrition is a Django-based web application prototype intended to support child nutrition surveillance. It organizes administrative geography, households, children, growth measurements, nutrition programmes and referrals. The project is divided into four Django apps: `accounts`, `geography`, `households` and `nutrition`. It uses SQLite for development and a Bootstrap/AdminLTE interface. This report summarizes the system's purpose, requirements, design, implementation and observed interface. Further workflow testing is needed before operational use.

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Chapter 1

Introduction

1.1 Background and Problem

Community health workers and district officers need structured records of household conditions, child growth measurements, programme enrolment and referrals. Rwanda Nutrition aims to centralize these records in a relational database and make them accessible through a browser-based interface.

1.2 Objective

To design and develop a database-driven Django web application for recording and organizing child nutrition surveillance information in Rwanda.

1.3 Specific Objectives

- Organize provinces, districts, sectors and health facilities.
- Register households and children.
- Record growth measurements, programme enrolments and referrals.
- Provide user accounts and a dashboard for reviewing key information.

Chapter 2

Requirements and Technologies

2.1 Main Functions

- Create, view, update and delete household and child records.
- Maintain geography and health-facility records.
- Record measurements, programmes, enrolments and referrals.
- Provide login and administrative record management.

2.2 Technology Stack

Technology	**Purpose**
Python and Django	Application logic, routing, views, models, forms and authentication.
SQLite	Development database.
HTML, CSS and JavaScript	Page structure, styling and interactions.
Bootstrap and AdminLTE	Responsive interface and dashboard.
Chart.js and DataTables	Visualization and enhanced tables where used.

Chapter 3

System Design

3.1 Application Modules

- `accounts`: login/logout and profile/role information.
- `geography`: provinces, districts, sectors and health facilities.
- `households`: household and child records.
- `nutrition`: measurements, programmes, enrolments, referrals and dashboard.

3.2 Database Relationships

- One province has many districts; one district has many sectors.
- One sector can contain many households and health facilities.
- One household can have many children.
- One child can have many growth measurements and referrals.
- Children and programmes are linked through an enrolment model.
- A user has a one-to-one relationship with a community health worker profile.

The project includes an ER diagram in its `erd` directory. The diagram should be supplied with the source code as part of the submission.

3.3 Application Flow

A browser request is matched to a Django URL pattern, routed to a view, processed using models and forms, and rendered through a template. Shared templates and AdminLTE assets provide a consistent interface.

Chapter 4

Implementation and Interface

The project contains four Django apps, shared templates, static assets, migrations, a development SQLite database and an administrative interface. The following figures are screenshots supplied by Group 4 from the application running locally at `127.0.0.1:8000`.

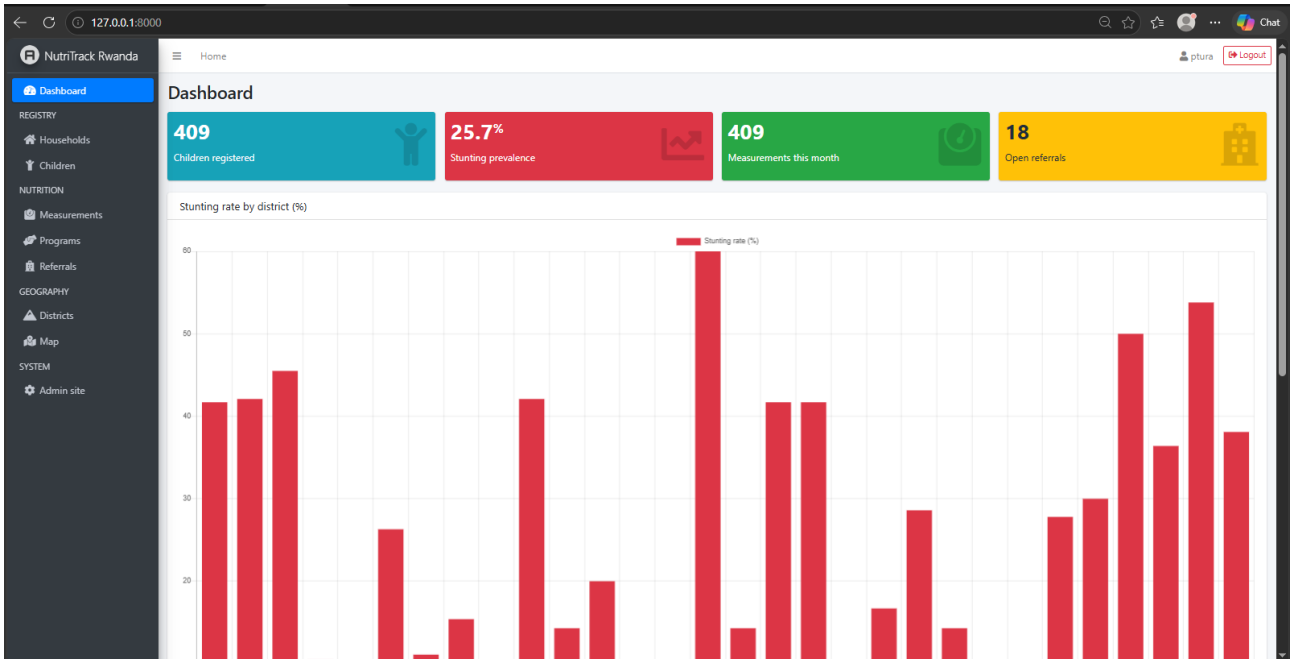


Figure 4.1: Dashboard overview with registration totals, stunting prevalence, monthly measurements, referrals and district chart.

4.1 Registry Screens

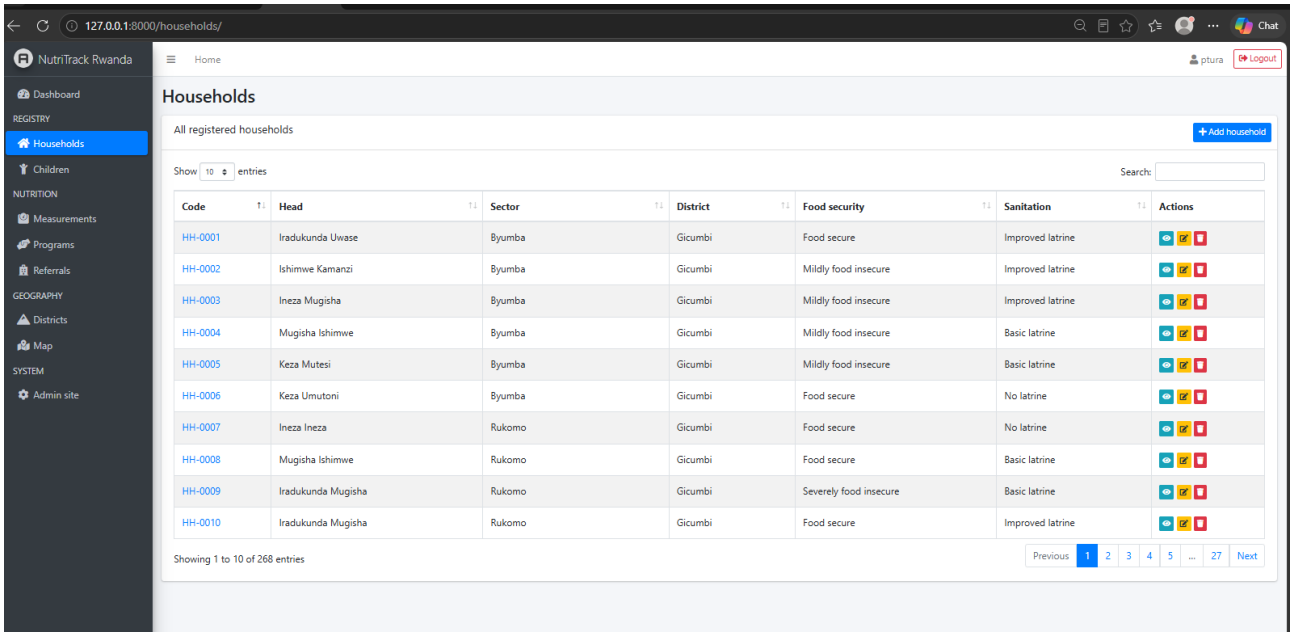


Figure 4.2: Household registry.

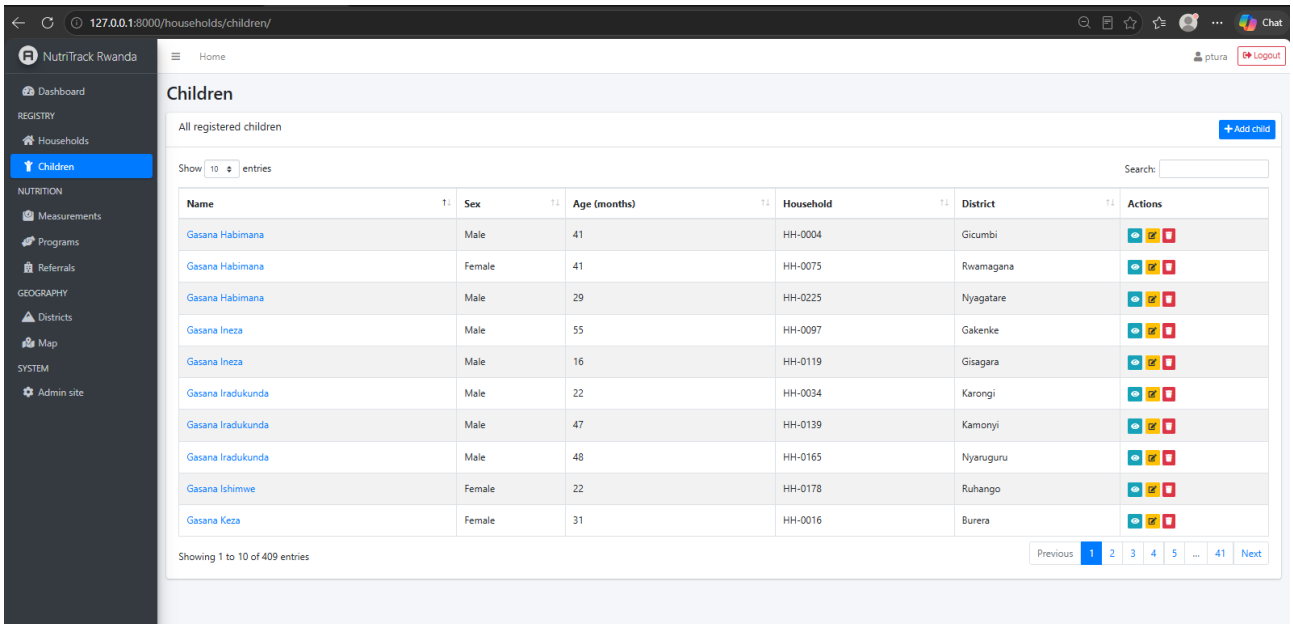


Figure 4.3: Child registry.

4.2 Django Administration

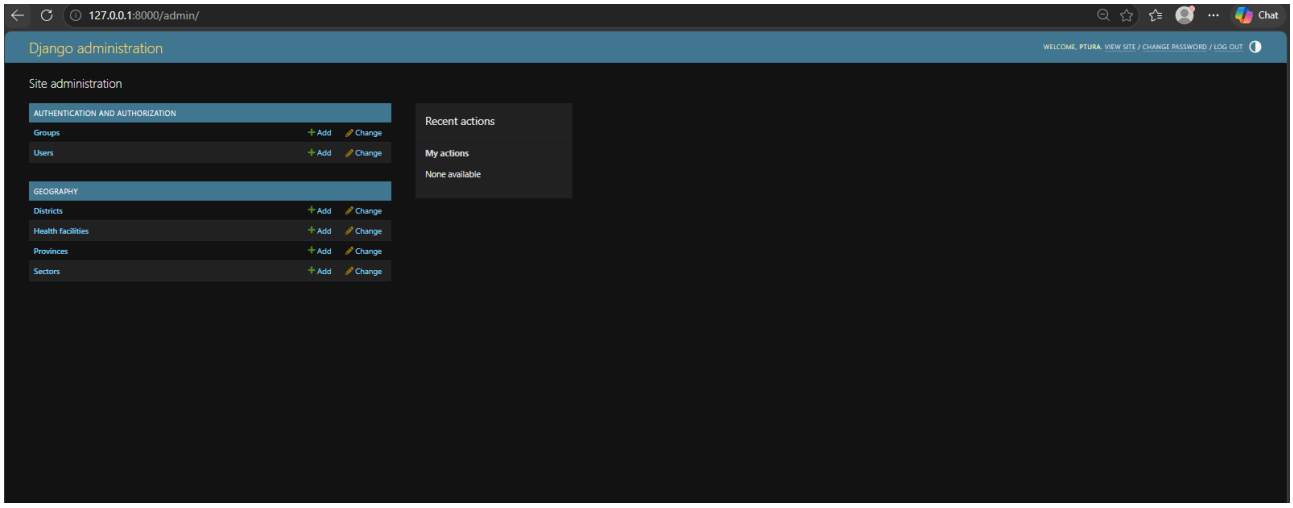


Figure 4.4: Django administration interface.

Chapter 5

Testing and Results

The screenshots show that the dashboard, household registry, child registry and Django administration page rendered in the local browser. Screenshots alone do not confirm that every form, delete confirmation, validation rule, permission and nutrition calculation works correctly.

Check	**Evidence / result**
Dashboard	Observed in supplied screenshot.
Household registry	Observed in supplied screenshot.
Child registry	Observed in supplied screenshot.
Django administration	Observed in supplied screenshot.
CRUD workflows, validation and access control	Require separate functional testing.

Chapter 6

Limitations and Recommendations

- The application is a prototype and needs further functional testing.
- SQLite is configured for development; deployment requirements should guide the production database choice.
- Protect child and household data with appropriate permissions, privacy safeguards and secure deployment.
- Validate nutrition indicators and any high-risk classifications against authoritative public-health guidance.
- Keep credentials and secret keys out of source control, and plan regular backups.

Chapter 7

Conclusion

Rwanda Nutrition provides a modular Django foundation for organizing child nutrition surveillance information. Its modules cover user profiles, geography, household and child registration, and nutrition activities. The supplied screenshots document the dashboard, registries and Django administration interface. Functional testing and security review remain important before operational use.

References

1. Django Software Foundation, *Django documentation*: https://docs.djangoproject.com/.
2. AdminLTE documentation: https://adminlte.io/docs/.
3. Rwanda Nutrition project source code and README supplied by Group 4. https://github.com/Be9321/RWANDA_NUTRITION.git/.